

Lecture 02 - Evolution of Internet (Notes)

1. What is the Internet?

The Internet is a global network of interconnected computers that communicate using standard networking protocols. It allows users to share information, communicate, and access online services.

2. Binary Language

Computers understand only binary language (0 and 1).

0 = OFF 1 = ON

Diagram

User → Text → Binary (0,1) → Computer Processing → Output

3. ASCII

ASCII (American Standard Code for Information Interchange) assigns numeric values to English characters.

Character	ASCII
A	65
B	66
a	97
Space	32

4. UTF Encoding

UTF-8, UTF-16 and UTF-32 are Unicode encoding formats used to represent characters from different languages.

Encoding	Size	Usage
UTF-8	1-4 Bytes	Most common on Internet
UTF-16	2-4 Bytes	Windows, Java
UTF-32	4 Bytes	Fixed length representation

5. Programming Languages

Machine Language → Assembly Language → High Level Language (C, C++, Java, Python)

6. Analog vs Digital Signals

Analog Signal: Continuous values.

Digital Signal: Discrete values (0 and 1).

7. Evolution of Internet

During the Cold War, the USA developed ARPANET to create a reliable communication network. ARPANET became the foundation of the modern Internet.

Timeline

1950s → Computers

1960s → ARPANET
1983 → TCP/IP
1990 → World Wide Web

8. Switching Techniques

Circuit Switching

Dedicated communication path between sender and receiver.

Diagram: Sender ===== Receiver

Message Switching

Entire message is stored and forwarded through intermediate nodes.

Diagram: Sender → Node → Node → Receiver

Packet Switching

Message is divided into small packets that travel independently and are reassembled at the destination.

Diagram:

Packet1 ↘

Packet2 → Network → Receiver

Packet3 ↗

9. Important Exam Questions

- Define Internet.
- What is Binary Language?
- What is ASCII?
- Difference between UTF-8, UTF-16 and UTF-32.
- Difference between Analog and Digital Signals.
- Explain ARPANET.
- Explain Circuit Switching.
- Explain Message Switching.
- Explain Packet Switching.

10. Summary

The Internet evolved from ARPANET. Computers communicate using binary language. ASCII and Unicode represent characters, while packet switching is the primary communication technique used on modern networks.